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Transfers - Decision Rules

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Descriptionâ

Decision Rules provide the final recommendation to whether a financial institution should accept, review, or decline a given transaction. This recommendation is sent via the API response of a request evoked by a scoring endpoint ([Transfer Initiation]). *Decision Rules* take into consideration the following:

- Rules in preceding rules projects in the workflow that triggered.
- The API field `payment_sub_type` to apply specific conditions to credit transfers, direct debits and instant transfers. For example, to only accept or decline events with the `payment_sub_type` field equal to `instant_transfers` (i.e. never send them to review).

- The score set by the model. By default, the solution does not provide a standard model in the workflow, and thus, the default score is 0.

The *Decision Rules* apply the following logic:

- If the score given by a model is above the review or decline thresholds, the transfer is marked as `review` or `declined`, respectively.
- All the rules in the Pulse workflow contribute to the `risk_level` outcome. This outcome is then taken into account by the *Decision Rules* to define the final recommendation using the `decision` outcome. If the previous rules mark the `risk_level` outcome as:
    - `low`: the *Decision Rules* mark the event as `approved`.
    - `medium`: the *Decision Rules* mark the event as `review`.
    - `high`: the *Decision Rules* mark the event as `decline`.
- The `enforced_risk_level` field is taken into account by the *Decision Rules*, and might differ from the `risk_level` outcome if more than one rule is triggered. The purpose of this field is to make sure that the risk level taken into account is the one with the highest priority.
- The *Decision Rules* also assure that if the `payment_sub_type` API field is `instant transfer`, the event is never sent to review. Instead, it is only `declined` or `accepted`, depending on the rules that triggered and the score given by the model.

Check the [Understand How Rules Work] page for more information about this rule logic.

About the review recommendationâ

Feedzai's customers can use the review outcomes according to the business needs. Here are a few examples of how to leverage this outcome value:

- Exclude the transfer from being automatically sent in the daily batch until the analyst takes the decision to manually include it (by labeling as not fraud).
- Include the transfer anyway, and continue its processing unless the analysts mark it as fraud before the cut-off time.

About the rules in this rules projectâ

There are two rules for each final recommendation decision (i.e. accept, review, decline), one rule is associated with the outcome generated by previous rules whereas the other rule is associated with the score generated by the model.

The order of the rules is also important to generate the right recommendation. The first pair of rules set the `decision` outcome as `decline`, the second pair generate `review`, and the last pair generate `accept`.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-DR1

Comment

Approve a Transfer if a whitelist rule was triggered

Actions

- [Transfers-DR1]

Variables

N/A

Condition

```
(event.enforced_risk_level ?? 'none') == 'low'
```

Explanation

[Transfers-DR7] - Approve - Classified as having a low risk of fraud by a whitelist rule.

Mark As

The outcome of this rule is: **approve**

Transfers-DR2

Comment

Decline an Instant Transfer if model score is above the threshold

Actions

- [Transfers-DR2]
- Alert

Variables

```
 HIGH_SCORE_THRESHOLD = transfers__rules_thresholds["model_high_score"].threshold;
LOW_SCORE_THRESHOLD = transfers__rules_thresholds["model_low_score"].threshold;

score = event.fraud_score * 1000;
```

Condition

```
 (score ?? 0) >= HIGH_SCORE_THRESHOLD
  or
((score ?? 0) > LOW_SCORE_THRESHOLD and event.payment_sub_type == 'instant_transfer' and
event.instant_transfers_review_outco == false)
```

Explanation

[Transfers-DR2] - Decline - Model score ({localvar.score}) was above the threshold: {localvar.HIGH_SCORE_THRESHOLD} ({localvar.LOW_SCORE_THRESHOLD} for Instant Transfers).

Mark As

The outcome of this rule is: **decline**

Transfers-DR3

Comment

Decline a Transfer if classified as high risk by previous rules

Actions

- [Transfers-DR3]
- Alert

Variables

N/A

Condition

```
(event.enforced_risk_level ?? event.risk_level)  == 'high'
  or
((event.enforced_risk_level ?? event.risk_level) == 'medium' and event.payment_sub_type ==
'instant_transfer' and event.instant_transfers_review_outco == false)
```

Explanation

[Transfers-DR3] - Decline - Classified as having a high risk of fraud by the rules.

Mark As

The outcome of this rule is: **decline**

Transfers-DR4

Comment

Mark a Transfer to Pending Review if model score is between medium and high risky thresholds

Actions

- Alert
- [Transfers-DR4]

Variables

```
HIGH_SCORE_THRESHOLD = transfers__rules_thresholds["model_high_score"].threshold;
LOW_SCORE_THRESHOLD = transfers__rules_thresholds["model_low_score"].threshold;

score = event.fraud_score * 1000;
```

Condition

```
(score ?? 0) > LOW_SCORE_THRESHOLD
```

Explanation

[Transfers-DR4] - Review - Model score ({localvar.score}) was between the review and risky thresholds ({localvar.LOW_SCORE_THRESHOLD} - {localvar.HIGH_SCORE_THRESHOLD})

Mark As

The outcome of this rule is: **review**

Transfers-DR5

Comment

Mark a Transfer to Pending Review if classified as medium risk by previous rules

Actions

- Alert
- [Transfers-DR5]

Variables

N/A

Condition

```
(event.enforced_risk_level ?? event.risk_level) == 'medium'
```

Explanation

[Transfers-DR5] - Review - Classified as having a medium risk of fraud by the rules.

Mark As

The outcome of this rule is: **review**

Transfers-DR6

Comment

Approve a Transfer if model score is below the threshold

Actions

- [Transfers-DR6]

Variables

```
 LOW_SCORE_THRESHOLD = transfers__rules_thresholds["model_low_score"].threshold;

score = event.fraud_score * 1000;
```

Condition

```
(score ?? 0) < LOW_SCORE_THRESHOLD
```

Explanation

[Transfers-DR6] - Approve - Model score ({localvar.score}) was below the threshold: {localvar.LOW_SCORE_THRESHOLD}

Mark As

The outcome of this rule is: **approve**

Transfers-DR7

Comment

Approve a Transfer if classified as low risk by previous rules

Actions

- [Transfers-DR7]

Variables

N/A

Condition

```
(event.enforced_risk_level ?? event.risk_level) == 'low'
```

Explanation

[Transfers-DR7] - Approve - Classified as having a low risk of fraud by the rules.

Mark As

The outcome of this rule is: **approve**